

Unsupervised Learning

Cluster Analysis

What is a Cluster?

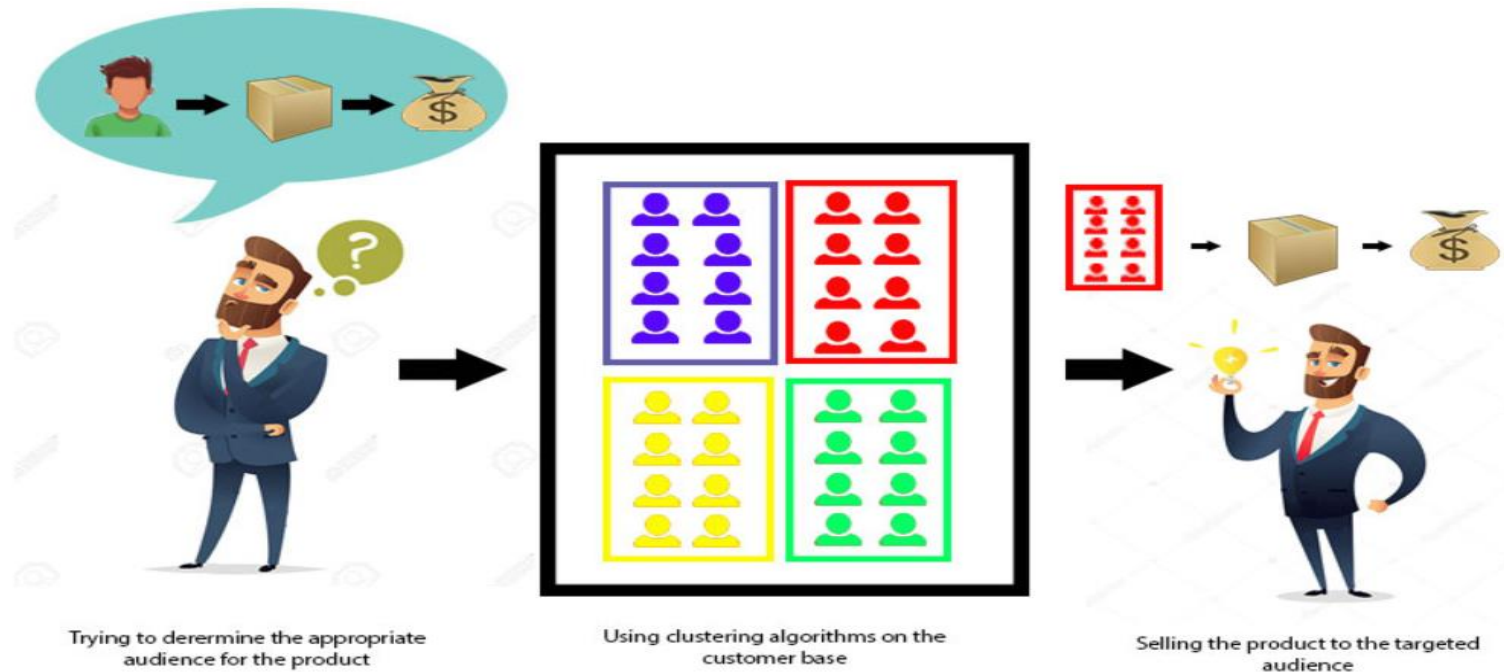
- A cluster is a subset of similar objects
- A subset of objects such that the distance between any of the two objects in the cluster is less than the distance between any object in the cluster and any object that is not located inside it.
- A connected region of a multidimensional space with a comparatively high density of objects.

Clustering in Data Mining

Clustering is an unsupervised Machine Learning-based Algorithm that comprises a group of data points into clusters so that the objects belong to the same group.

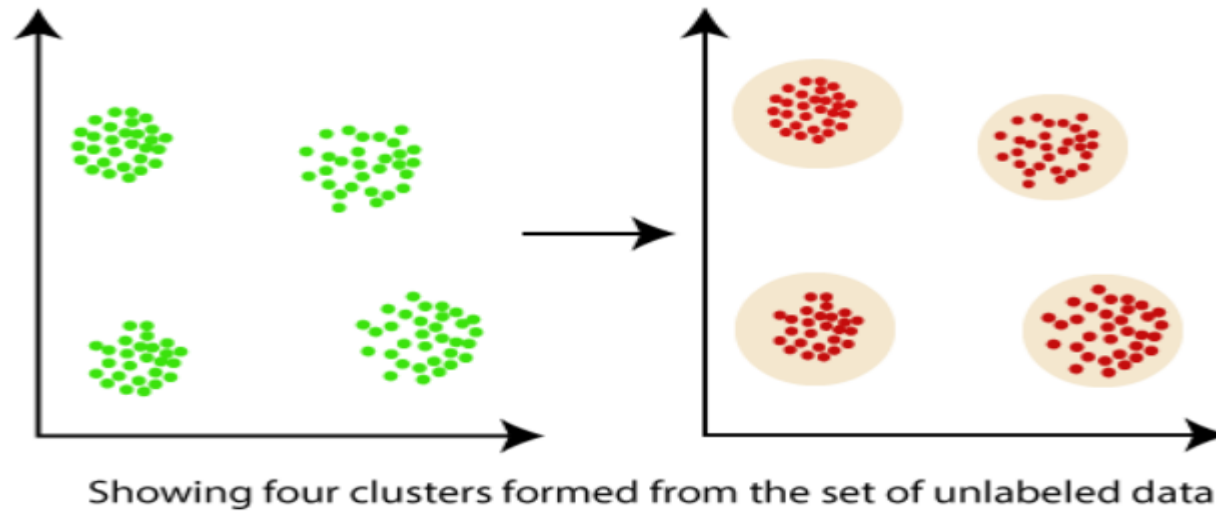
Clustering in Data Mining Continue

Clustering helps to splits data into several subsets. Each of these subsets contains data similar to each other, and these subsets are called clusters. Now that the data from our customer base is divided into clusters, we can make an informed decision about who we think is best suited for this product.



Clustering in Data Mining Continue

Let's understand this with an example, suppose we are a **market manager**, and we have a new **tempting product to sell**. We are sure that the product would bring enormous profit, as long as it is sold to the right people. So, how can we tell who is best suited for the product from our company's huge customer base?



Clustering in Data Mining Continue

- Clustering, falling under the category of **unsupervised machine learning**, is one of the problems that machine learning algorithms solve.
- Clustering only **utilizes input data, to determine patterns, anomalies, or similarities in its input data.**

A good clustering algorithm aims to obtain clusters whose:

- The intra-cluster similarities are high; it implies that the data present inside the cluster is similar to one another.
- The inter-cluster similarity is low, and it means each cluster holds data that is not similar to other data.

Types of Clustering

1) Partitioning Clustering

How it works: Divides the dataset into a predefined number of clusters (K). The algorithm assigns each point to the closest cluster center (centroid) and iteratively improves until the clusters are well-formed.

Common algorithm: K-Means Clustering.

Best for: Fast processing on large datasets with clearly separated, spherical clusters.

Types of Clustering

2) Hierarchical Clustering

How it works: Builds a tree-like structure of clusters (called a dendrogram). It can work **Agglomeratively** (starts with individual points and merges them) or **Divisively** (starts with one big cluster and breaks it down).

Common algorithms: Agglomerative Clustering, BIRCH.

Best for: Datasets with an inherent nested structure (like taxonomy or family trees) where you want to visualize data at different levels of granularity.

Types of Clustering

3) Density-Based Clustering

How it works: Groups data points based on areas of high concentration (density) separated by areas of low concentration. Points in sparse regions are treated as noise or outliers.

Common algorithms: DBSCAN, HDBSCAN.

Best for: Finding clusters of arbitrary, irregular shapes and handling datasets with significant noise

Types of Clustering

4) Distribution-Based Clustering

How it works: Assumes data is composed of distributions, such as Gaussian (normal) distributions. It groups data based on the probability that they belong to the same statistical distribution.

Common algorithm: Gaussian Mixture Models (GMM).

Best for: Real-world datasets where clusters might overlap or vary in size and shape.

Types of Clustering

5) Grid-Based Clustering

How it works: Divides the data space into a finite grid of cells (like a chessboard) and performs all operations on the grid rather than directly on the data points.

Common algorithms: STING, CLIQUE.

Best for: Spatial data, mapping, and handling massive datasets where computational speed is a priority.

Clustering Assignment Methods

Beyond the structural types, clustering methods are broadly distinguished by how points are assigned to groups:

- **Hard Clustering (Exclusive)**: Each data point belongs to exactly one cluster with no overlap. A point cannot be in two groups simultaneously (e.g., K-Means).
- **Soft Clustering (Overlapping)**: A data point can belong to multiple clusters with varying probabilities or degrees of membership (e.g., Gaussian Mixture Models).

Applications of cluster analysis in data mining

- In many applications, clustering analysis is widely used, such as data analysis, market research, pattern recognition, and image processing.
- It assists marketers to find different groups in their client base and based on the purchasing patterns. They can characterize their customer groups.
- It helps in allocating documents on the internet for data discovery.
- Clustering is also used in tracking applications such as detection of credit card fraud.
- As a data mining function, cluster analysis serves as a tool to gain insight into the distribution of data to analyze the characteristics of each cluster.
- In terms of biology, It can be used to determine plant and animal taxonomies, categorization of genes with the same functionalities and gain insight into structure inherent to populations.
- It helps in the identification of areas of similar land that are used in an earth observation database and the identification of house groups in a city according to house type, value, and geographical location.

Why is clustering used in data mining?

- Clustering analysis has been an evolving problem in data mining due to its variety of applications.
- It's **comprehensive** use in a broad range of applications, including
 - image processing, computational biology, mobile communication, medicine, and economics

K-Means Clustering Example

A1(1,1),A2(2,1),A3(4,3),A4(5,4)

K=2

<u>S.No.</u>	X1	Y1
A1	1	1
A2	2	1
A3	4	3
A4	5	4

Step 1

Initial Seed

Seed 1 A2 (2,1)

Seed 2 A4 (5,4)

Apply Euclidean Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Step 2

A1

$$d(A1, \text{Seed 1}) = \sqrt{(2 - 1)^2 + (1 - 1)^2} = 1 \quad \text{Smallest}$$

$$d(A1, \text{Seed 2}) = \sqrt{(5 - 1)^2 + (4 - 1)^2} = 5$$

A1 ∈ Cluster 1

A2

$$d(A2, \text{Seed 1}) = 0 \quad \text{Smallest}$$

$$d(A2, \text{Seed 2}) = \sqrt{(5 - 2)^2 + (4 - 1)^2} = 4.24$$

A2 ∈ Cluster 1

A3

$$d(A3, \text{Seed 1}) = \sqrt{(2 - 4)^2 + (1 - 3)^2} = 2.82$$

$$d(A3, \text{Seed 2}) = \sqrt{(5 - 4)^2 + (4 - 3)^2} = 1.41 \quad \text{Smallest}$$

A3 ∈ Cluster 2

A4

$$d(A4, \text{Seed 1}) = \sqrt{(2 - 5)^2 + (1 - 4)^2} = 4.24$$

$$d(A4, \text{Seed 2}) = \sqrt{(5 - 5)^2 + (4 - 4)^2} = 0 \quad \text{Smallest}$$

A4 ∈ Cluster 2

A1(1,1), A2(2,1), A3(4,3), A4(5,4)

K=2

<u>S.No.</u>	X1	Y1
A1	1	1
A2	2	1
A3	4	3
A4	5	4

Step 1

Initial Seed

Seed 1 A2 (2,1)

Seed 2 A4 (5,4)

Step 3

New Cluster are

Cluster 1 {A1,A2}

Cluster 2 {A3,A4}

Centroid

$$C1 = \frac{1+2}{2}, \frac{1+1}{2} = 1.5, 1$$

$$C2 = \frac{4+5}{2}, \frac{3+4}{2} = 4.5, 3.5$$

A1(1,1),A2(2,1),A3(4,3),A4(5,4)

K=2

<u>S.No.</u>	X1	Y1
A1	1	1
A2	2	1
A3	4	3
A4	5	4

Step 1

Initial Seed

Seed 1 A2 (2,1)

Seed 2 A4 (5,4)

K-Means Clustering – Solved Example

- Suppose that the data mining task is to cluster points into three clusters,
- where the points are
- $A1(2, 10)$, $A2(2, 5)$, $A3(8, 4)$, $B1(5, 8)$, $B2(7, 5)$, $B3(6, 4)$, $C1(1, 2)$, $C2(4, 9)$.
- The distance function is Euclidean distance.
- Suppose initially we assign $A1$, $B1$, and $C1$ as the center of each cluster, respectively.

K-Means Clustering – Solved Example

Initial Centroids:

A1: (2, 10)

B1: (5, 8)

C1: (1, 2)

Data Points			Distance to						Cluster	New Cluster
X1	Y1	X2	Y2	Distance to						
A1	2	10	2	10	5	8	1	2		
A2	2	5								
A3	8	4								
B1	5	8								
B2	7	5								
B3	6	4								
C1	1	2								
C2	4	9								

K-Means Clustering – Solved Example



Initial Centroids:

A1: (2, 10)

B1: (5, 8)

C1: (1, 2)

Data Points X1 Y1			X2	Y2	Distance to				Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0.00		3.61		8.06		1	
A2	2	5	5.00		4.24		3.16		3	
A3	8	4	8.49		5.00		7.28		2	
B1	5	8	3.61		0.00		7.21		2	
B2	7	5	7.07		3.61		6.71		2	
B3	6	4	7.21		4.12		5.39		2	
C1	1	2	8.06		7.21		0.00		3	
C2	4	9	2.24		1.41		7.62		2	

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

2nd Method K-Means Clustering Example

Step 1: Take mean value

Step 2: Find nearest number of mean and put in cluster

Step 3: Repeat Step 1 and Step 2 until we get same mean

Data = {2, 3, 4, 10, 11, 12, 20, 25, 30}

Where K = 02

M1=4

M2= 12

Iteration 1

C1 = {2,3,4}

M1 = (2+3+4)/3

M1 = 9/3

M1 = 3

C2 = {10,11,12,20,25,30}

M2 = (10+11+12+20+25+30)/6

M2 = 108/6

M2 = 18

Data = {2, 3, 4, 10, 11, 12, 20, 25, 30}

Where K = 02

M1 = 4

M2 = 12

Iteration 2

Data = {2, 3, 4, 10, 11, 12, 20, 25, 30}

M1 = 3

M2 = 18

C1 = {2,3,4,10}

M1 = (2+3+4+10)/4

M1 = 4.75

M1 = 5

C2 = {11,12,20,25,30}

M2 = (11+12+20+25+30)/5

M2 = 19.6

M2 = 20

Iteration 3

Data = {2, 3, 4, 10, 11, 12, 20, 25, 30}

M1=5

M2= 20

$C1=\{2,3,4,10,11,12\}$ $M1=(2+3+4+10+11+12)/6$ M1 =7	$C2=\{20,25,30\}$ $M2=(20+25+30)/3$ M2=25
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Iteration 4

Data = {2, 3, 4, 10, 11, 12, 20, 25, 30}

M1=7

M2= 25

$C1=\{2,3,4,10,11,12\}$ $M1=(2+3+4+10+11+12)/6$ M1=7	$C2=\{20,25,30\}$ $M2=(20+25+30)/3$ M2=25
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Thus we get same mean so new cluster will be

C1= {2,3,4,10,11,12} and C2={20,25,30}

Agglomerative clustering

- Agglomerative clustering works in a “bottom-up” manner. That is, each object is initially considered as a single-element cluster (leaf). At each step of the algorithm, the two clusters that are the most similar are combined into a new bigger cluster (nodes). This procedure is iterated until all points are member of just one single big cluster (root).

Steps of Agglomerative Hierarchical Clustering

We'll follow the steps below to perform agglomerative hierarchical clustering :

1. Preparing the data
2. Computing (dis)similarity information between every pair of objects in the data set.
3. Using linkage function to group objects into hierarchical cluster tree, based on the distance information generated at step 1. Objects/clusters that are in close proximity are linked together using the linkage function.
4. Determining where to cut the hierarchical tree into clusters. This creates a partition of the data.

Linkage

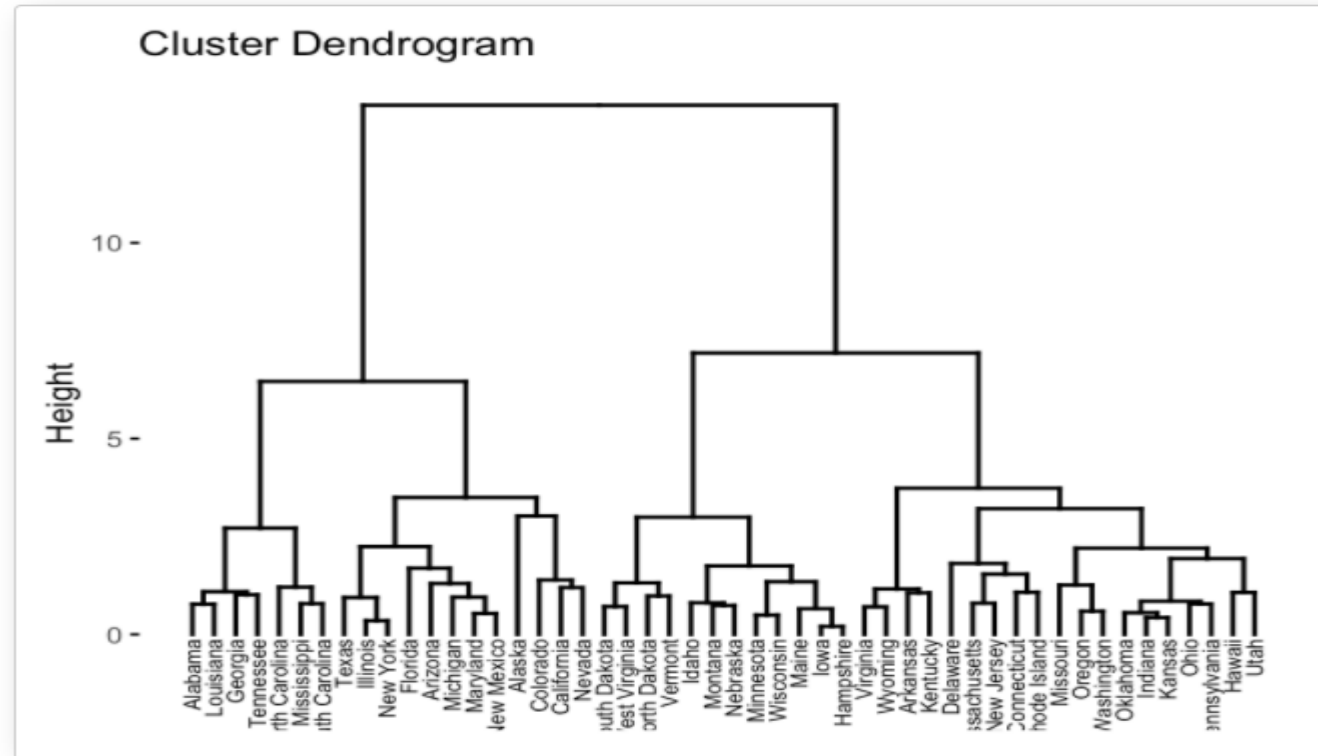
The linkage function takes the distance information and groups pairs of objects into clusters based on their similarity. Next, these newly formed clusters are linked to each other to create bigger clusters. This process is iterated until all the objects in the original data set are linked together in a hierarchical tree.

Method

- **Maximum or complete linkage:** The distance between two clusters is defined as the maximum value of all pairwise distances between the elements in cluster 1 and the elements in cluster 2. It tends to produce more compact clusters.
- **Minimum or single linkage:** The distance between two clusters is defined as the minimum value of all pairwise distances between the elements in cluster 1 and the elements in cluster 2. It tends to produce long, “loose” clusters.
- **Mean or average linkage:** The distance between two clusters is defined as the average distance between the elements in cluster 1 and the elements in cluster 2.
- **Centroid linkage:** The distance between two clusters is defined as the distance between the centroid for cluster 1 (a mean vector of length p variables) and the centroid for cluster 2.
- **Ward's minimum variance method:** It minimizes the total within-cluster variance. At each step the pair of clusters with minimum between-cluster distance are merged

Dendrogram

Dendrograms correspond to the graphical representation of the hierarchical tree.



Dendrogram Continue

- In the dendrogram, each leaf corresponds to one object. As we move up the tree, objects that are similar to each other are combined into branches, which are themselves fused at a higher height.
- The height of the fusion, provided on the vertical axis, indicates the (dis)similarity/distance between two objects/clusters. The higher the height of the fusion, the less similar the objects are. This height is known as the *cophenetic distance* between the two objects.

Example: Minimum - Single Linkage

Item	A	B	C	D	E
A	0				
B	5	0			
C	2	1	0		
D	3	3	6	0	
E	1	2	2	3	0

It is found that smallest value found at (EA) → 1

It is found that smallest value found at (CB) → 1

$$BC \longrightarrow EA = \min[BE, BA, CE, CA]$$

$$= \min [2, 5, 2, 2]$$

$$= 2$$

$$D \longrightarrow EA = \min[ED, AD]$$

$$= \min[3, 3]$$

$$= 3$$

$$D \longrightarrow BC = \min[BD, CD]$$

$$= \min [3, 6]$$

$$= 3$$

Item	A	B	C	D	E
A	0				
B	5	0			
C	2	1	0		
D	3	3	6	0	
E	1	2	2	3	0

Item	EA	BC	D
EA	0		
BC	2	0	
D	3	3	0

Now check again, you will find minimum value i.e. 02, it exist at Pair (EA) and (BC) $\longrightarrow$ 02

Item	(EA),(BC)	D
(EA),(BC)	0	
D	3	0

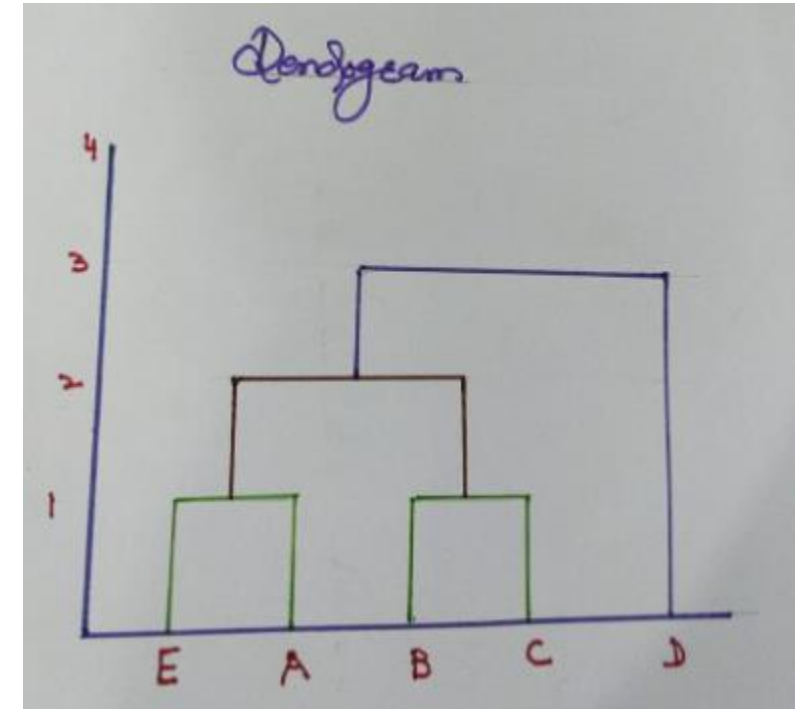
Item	A	B	C	D	E
A	0				
B	5	0			
C	2	1	0		
D	3	3	6	0	
E	1	2	2	3	0

$(EA),(BC) \longrightarrow D = \min [ED, AD, BD, CD]$
 $= \min[3, 3, 3, 6]$
 $= 3$

We found minimum value at (EA),(BC) and D $\longrightarrow$ 03

We found that:

1	$(E,A) \longrightarrow 1$
2	$(BC) \longrightarrow 1$
3	$(EA) \text{ and } (BC) \longrightarrow 02$
4	$(EA),(BC) \text{ and } D \longrightarrow 03$



Example: Maximum - Complete Linkage

Its slow :: complicated :: repeatable :: not suited for big data sets.

Item	A	B	C	D	E
A	0				
B	5	0			
C	2	1	0		
D	3	3	6	0	
E	1	2	2	3	0

Complete Link Clustering:

Considers Max of all distances. Leads to many small clusters.

It is found that smallest value found at (EA) → 1

It is found that smallest value found at (CB) → 1

$$BC \longrightarrow EA = \max[BE, BA, CE, CA]$$

$$= \max [2, 5, 2, 2]$$

$$= 5$$

$$D \longrightarrow EA = \max[ED, AD]$$

$$= \max[3, 3]$$

$$= 3$$

$$D \longrightarrow BC = \max[BD, CD]$$

$$= \max [3, 6]$$

$$= 6$$

Item	A	B	C	D	E
A	0				
B	5	0			
C	2	1	0		
D	3	3	6	0	
E	1	2	2	3	0

Item	EA	BC	D
EA	0		
BC	5	0	
D	3	6	0

Now check again, you will find minimum value i.e. 03, it exist at Pair (EA) and D $\longrightarrow$ 3

Item	(EAD)	(BC)
(EAD)	0	
(BC)	6	0

Item	A	B	C	D	E
A	0				
B	5	0			
C	2	1	0		
D	3	3	6	0	
E	1	2	2	3	0

$$\begin{aligned}
 (EAD) \longrightarrow (BC) &= \max [EB, EC, AB, AC, DB, DC] \\
 &= \max[3, 3, 3, 6] \\
 &= 6
 \end{aligned}$$

We found minimum value at (EAD) and (BC) $\longrightarrow 6$